Problem H2: Electroweak Mixing of $W^{(1,2,3)}$ and B

Parker — PHYS 7502

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Problem Statement

H2. Give the relation that describes the "mixing" of the "fundamental" $\{W^1, W^2, W^3\}$ vector boson triplet and B vector boson of the electroweak theory, which get transformed in terms of the weak mixing angle into the usual 4 electroweak bosons we observe (W^+, W^-, Z^0, γ).

Deliverables

There are two relations that together do the full $4 \to 4$ transformation:
1. The **charged** mixing: $\{W^1, W^2\} \to \{W^+, W^-\}$.
2. The **neutral** mixing (the one with θ_W): $\{W^3, B\} \to \{\gamma, Z\}$.

This writeup states both and proves each. All references are to Griffiths, *Introduction to Elementary Particles* (2nd ed.), Ch. 9 §9.7.

Setup — the GWS Lagrangian (Griffiths Eq. 9.145)

Setup

The electroweak gauge group $SU(2)_L \otimes U(1)$ has $3 + 1 = 4$ generators, hence 4 gauge fields:

$$\underbrace{W^1_\mu,\ W^2_\mu,\ W^3_\mu}_{SU(2)_L \text{ triplet (coupling } g_W)} \qquad \text{and} \qquad \underbrace{B_\mu}_{U(1) \text{ singlet (coupling } g')} .$$

The GWS interaction Lagrangian is (Griffiths Eq. 9.145)

$$\mathcal{L}^{\text{GWS}}_{\text{int}} \;=\; -i\Big[\, g_W\, \vec{j}_\mu \cdot \vec{W}^\mu \;+\; \tfrac{g'}{2}\, j^Y_\mu\, B^\mu \,\Big] ,$$

where $\vec{j} \cdot \vec{W} = j^1 W^1 + j^2 W^2 + j^3 W^3$ is the dot product in weak isospin space, and j^Y is the weak hypercharge current with $j^Y = 2j^{\text{em}} - 2j^3$ (Griffiths Eq. 9.143; this is just Gell-Mann–Nishijima $Q = T_3 + Y/2$ at the current level).

The two relations we need to prove repackage these four gauge fields into the four physical bosons. The two sectors decouple: charged (W^1, W^2) and neutral (W^3, B) mix among themselves but not across.

Relation 1 — Charged sector $\{W^1, W^2\} \to \{W^+, W^-\}$

The Relation

Griffiths Eq. 9.148:

$$\boxed{W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \pm i\, W_\mu^2)\,.}$$

Equivalently, $W_\mu^1 = \frac{1}{\sqrt{2}}(W_\mu^+ + W_\mu^-)$ and $W_\mu^2 = \frac{i}{\sqrt{2}}(W_\mu^+ - W_\mu^-)$.

Proof

We prove this by demanding two things:

1. **The charged piece of the Lagrangian must be expressible in terms of physical, electrically-charged fields $W^\pm$.**
2. **The resulting vertex factor for $e^- \to \nu_e + W^-$ must reduce to the standard $V-A$ form** $-i(g_W/\sqrt{2})\,\gamma^\mu(1-\gamma^5)/2$ (Griffiths Eq. 9.5), which is the experimental input.

The unique combination satisfying both is Eq. 9.148.

Step 1: Why $W^\pm$ must be complex (not real)

Step

The W^+ has electric charge $+e$ and the W^- has $-e$ (they are charge-conjugate to each other). A real-valued field describes a particle that is its own antiparticle (e.g. the photon). Since $W^+ \neq W^-$ as particles, their fields must be *complex*, with $(W^+)^\dagger = W^-$.

Now W^1 and W^2 are real fields (the components of a Hermitian-basis $SU(2)$ gauge connection). The standard recipe for building a complex field from two real ones is $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$, $\phi^* = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2)$. This forces the form

$$W_\mu^+ = \tfrac{1}{\sqrt{2}}(W_\mu^1 + iW_\mu^2), \qquad W_\mu^- = (W_\mu^+)^\dagger = \tfrac{1}{\sqrt{2}}(W_\mu^1 - iW_\mu^2),$$

up to (i) overall normalization and (ii) which combination is labeled "+" versus "−". The labeling is fixed by experiment (which couples W^+ to j^-, not to j^+). The normalization is fixed in Step 2.

Step 2: The $\frac{1}{\sqrt{2}}$ factor is fixed by recovering V-A

Step

Start from the charged piece of the GWS Lagrangian (Eq. 9.145, isolating the W^1 and W^2 terms):

$$\mathcal{L}_{\rm CC} \;=\; -\,i\,g_W\,(j^1_\mu W^{1\mu} + j^2_\mu W^{2\mu}).$$

Plug in $W^1 = \frac{1}{\sqrt{2}}(W^+ + W^-)$, $W^2 = \frac{i}{\sqrt{2}}(W^+ - W^-)$:

$$\begin{aligned} j^1_\mu W^{1\mu} + j^2_\mu W^{2\mu} &= j^1 \cdot \tfrac{1}{\sqrt{2}}(W^+ + W^-) + j^2 \cdot \tfrac{i}{\sqrt{2}}(W^+ - W^-) \\ &= \tfrac{1}{\sqrt{2}}\left[(j^1 + ij^2)\,W^+ + (j^1 - ij^2)\,W^-\right] \\ &= \tfrac{1}{\sqrt{2}}\left[j^+_\mu\,W^{+\mu} + j^-_\mu\,W^{-\mu}\right], \end{aligned} \tag{1}$$

where in the last line I defined $j^\pm_\mu \equiv j^1_\mu \pm i\,j^2_\mu$. Therefore

$$\mathcal{L}_{\rm CC} \;=\; -\,i\,\frac{g_W}{\sqrt{2}}\left[j^+_\mu W^{+\mu} + j^-_\mu W^{-\mu}\right].$$

The factor $\frac{1}{\sqrt{2}}$ that appears in front is exactly what fixes the normalization in the definition of $W^\pm$. Without it, the V-A vertex factor would have a "wrong" coefficient.

Step 3: Verify that V-A drops out

Step

The current j^-_μ that couples to W^+ in (1) is, by construction ($j^- = j^1 - ij^2$, where j^1, j^2 are the standard isospin currents $\bar\chi_L \gamma_\mu \tau^a \chi_L$ for $a = 1, 2$):

$$j^-_\mu \;=\; \bar e_L\,\gamma_\mu\,\nu_L \;=\; \bar e\,\gamma_\mu \tfrac{(1-\gamma^5)}{2}\,\nu.$$

Plugging into the Lagrangian, the $\bar e\,\nu\,W^+$ vertex picks up the factor

$$\boxed{-\,i\,\frac{g_W}{\sqrt{2}}\,\gamma^\mu\frac{(1-\gamma^5)}{2}} \quad \text{(Griffiths Eq. 9.150)},$$

which is exactly the $V - A$ vertex factor we started with at Eq. 9.5.
Therefore the definition $W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \pm iW^2_\mu)$ is the unique choice that simultaneously (i) makes $W^\pm$ a charge-eigenstate complex field and (ii) reproduces the experimentally-known V-A vertex factor. ■

Convention Note

Sign convention. Griffiths writes $W^{\pm} = \frac{1}{\sqrt{2}}(W^1 \pm iW^2)$. Thomson, Mann, and Halzen & Martin use the opposite sign $W^{\pm} = \frac{1}{\sqrt{2}}(W^1 \mp iW^2)$. This just flips which combination is called "+" vs. "−" and has no physical content, provided you're consistent within one textbook. Stay with Griffiths.

Relation 2 — Neutral sector $\{W^3, B\} \to \{\gamma, Z\}$

The Relation

Griffiths Eq. 9.151 (the heart of the question):

$$\boxed{\begin{aligned} A_\mu\,(\equiv \gamma) &= \;\; B_\mu \cos\theta_W \;+\; W^3_\mu \sin\theta_W, \\ Z_\mu &= -B_\mu \sin\theta_W \;+\; W^3_\mu \cos\theta_W, \end{aligned}}$$

where θ_W is the weak mixing (Weinberg) angle, fixed by

$$\boxed{\tan\theta_W = \frac{g'}{g_W}, \qquad g_W \sin\theta_W = g' \cos\theta_W = g_e} \quad \text{(Griffiths Eq. 9.154).}$$

Proof

The proof has two parts:

1. **The mixing is the most general orthogonal transformation of the two real, neutral fields W^3 and B that preserves canonical kinetic-term normalization.** That fixes the *form* of the rotation up to a single free angle θ_W.
2. **The angle θ_W is fixed by demanding that one of the rotated fields (A_μ) couple to electric charge with strength g_e, reproducing QED.** That fixes the *value* of the rotation (Eq. 9.154).

Step 1: The rotation form is forced by orthogonality

Step

W^3_μ and B_μ are two real, neutral fields with canonically normalized kinetic terms in the Lagrangian:

$$\mathcal{L}_{\text{kin}} \;\supset\; -\tfrac{1}{4}(F^{(W^3)}_{\mu\nu})^2 \;-\; \tfrac{1}{4}(F^{(B)}_{\mu\nu})^2.$$

Any change of basis $\binom{B}{W^3} \to \binom{A}{Z}$ that preserves canonical normalization must be *orthogonal*, i.e. a 2×2 matrix R with $R^T R = \mathbb{1}$. Up to discrete reflections (which are just relabelings), every 2×2 orthogonal matrix is a rotation by a single angle:

$$R(\theta_W) \;=\; \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix}.$$

Therefore the most general allowed mixing is

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} \;=\; R(\theta_W) \begin{pmatrix} B_\mu \\ W^3_\mu \end{pmatrix} \;=\; \begin{pmatrix} B_\mu \cos\theta_W + W^3_\mu \sin\theta_W \\ -B_\mu \sin\theta_W + W^3_\mu \cos\theta_W \end{pmatrix},$$

which is the *form* given in Eq. 9.151. **One free parameter θ_W remains.**

Step 2: Substitute the inverse rotation into the Lagrangian

Step

The inverse rotation (which exists because R is orthogonal, $R^{-1} = R^T$) is

$$\begin{pmatrix} B_\mu \\ W^3_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix}, \quad \Rightarrow \begin{aligned} B_\mu &= A_\mu \cos\theta_W - Z_\mu \sin\theta_W, \\ W^3_\mu &= A_\mu \sin\theta_W + Z_\mu \cos\theta_W. \end{aligned}$$

The neutral piece of the GWS Lagrangian (from Eq. 9.145) is

$$\mathcal{L}_{\text{neut}} \;=\; -\,i\Big[g_W\, j^3_\mu\, W^{3\mu} + \tfrac{g'}{2}\, j^Y_\mu\, B^\mu\Big].$$

Substituting:

$$\begin{aligned} \mathcal{L}_{\text{neut}} &= -\,i\Big[g_W\, j^3_\mu (A^\mu \sin\theta_W + Z^\mu \cos\theta_W) + \tfrac{g'}{2}\, j^Y_\mu\, (A^\mu \cos\theta_W - Z^\mu \sin\theta_W)\Big] \\ &= -\,i\, A^\mu \Big(g_W \sin\theta_W\, j^3_\mu + \tfrac{g'}{2} \cos\theta_W\, j^Y_\mu\Big) \;+\; (\text{terms in } Z^\mu). \end{aligned}$$

This matches Griffiths Eq. 9.152.

Step 3: Demand that the A^μ piece reproduce QED

Step

The QED interaction is $\mathcal{L}_{\text{QED}} = -i\, g_e\, j^{\text{em}}_\mu A^\mu$ (Griffiths Eq. 9.153). For our rotated Lagrangian to contain this at low energies, we require

$$g_W \sin\theta_W\, j^3_\mu \;+\; \tfrac{g'}{2}\cos\theta_W\, j^Y_\mu \;\stackrel{!}{=}\; g_e\, j^{\text{em}}_\mu .$$

Now use the Gell-Mann–Nishijima formula at the current level (Griffiths Eq. 9.143: $j^{\text{em}} = j^3 + \frac{1}{2}j^Y$):

$$g_W \sin\theta_W\, j^3 \;+\; \tfrac{g'}{2}\cos\theta_W\, j^Y \;=\; g_e\, j^3 \;+\; \tfrac{g_e}{2}\, j^Y .$$

Match the coefficients of j^3 and j^Y *separately* (since they are independent currents):

$$\text{coeff. of } j^3: \quad g_W \sin\theta_W \;=\; g_e,$$
$$\text{coeff. of } j^Y: \quad \tfrac{g'}{2}\cos\theta_W \;=\; \tfrac{g_e}{2} \;\Longleftrightarrow\; g'\cos\theta_W \;=\; g_e.$$

Combining these gives Griffiths Eq. 9.154:

$$\boxed{g_W \sin\theta_W \;=\; g'\cos\theta_W \;=\; g_e.}$$

Dividing the two halves of the equality eliminates g_e:

$$\boxed{\tan\theta_W \;=\; \frac{g'}{g_W}.}$$

This single equation pins down the angle θ_W in terms of the two underlying gauge couplings. ■

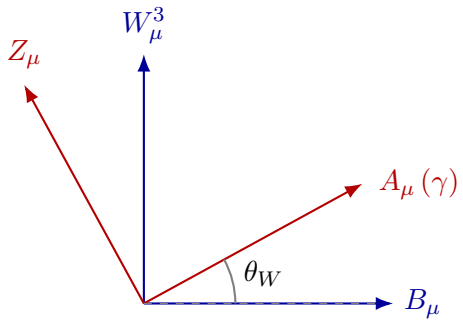
Why the proof works

Physical Insight

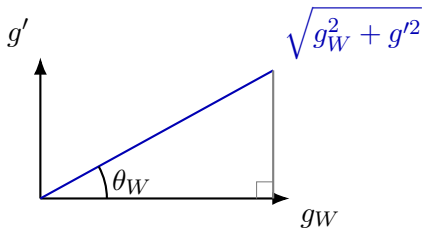
Logic of the proof, recapped.

- The *form* of the mixing (rotation by some angle) is forced by the kinematic structure of two real neutral fields with canonical kinetic terms (Step 1). No physics input yet.
- The *value* of the angle is forced by demanding that one rotated field be the QED photon (Step 2–3). This is the physics input: at low energies the theory must reduce to ordinary electromagnetism with coupling g_e.
- The Gell-Mann–Nishijima formula $Q = T_3 + Y/2$ (which is built into the theory by the choice of hypercharge assignments) is what allows the matching to succeed — it ensures j^{em} is exactly the right linear combination of j^3 and j^Y that the rotated Lagrangian produces.

Picture: the rotation in field space and the coupling triangle



Rotation in the neutral-field plane



Triangle of couplings

Brief: *why* does the mixing happen?

Intuition

The proof above shows the mixing is *consistent* with QED, but it doesn't say what *causes* it. Griffiths defers this to his §10.9 ("The Higgs Mechanism"), which is worth one summary paragraph here for completeness:
The Higgs vacuum expectation value $\langle\Phi\rangle = \frac{1}{\sqrt{2}}(0, v)^T$ is *electrically neutral* ($Q\langle\Phi\rangle = 0$, because of the way hypercharges are assigned). It carries weak isospin and weak hypercharge separately, so it is acted on non-trivially by T^1, T^2, and the combination $(T_3 - \frac{1}{2}Y)$ — but *not* by $Q = T_3 + \frac{1}{2}Y$. Plugging $\langle\Phi\rangle$ into the Higgs kinetic term $|D_\mu\Phi|^2$ generates a mass matrix for the gauge fields; in the (W^3, B) subspace this matrix is rank-1 with a zero eigenvalue. The eigenvector with zero mass is exactly $A_\mu = B_\mu \cos\theta_W + W^3_\mu \sin\theta_W$ (the photon, gauge field of the unbroken $U(1)_{\text{em}}$ generated by Q); the orthogonal massive eigenvector is $Z_\mu = -B_\mu \sin\theta_W + W^3_\mu \cos\theta_W$, with $\tan\theta_W = g'/g_W$ — exactly Eq. 9.151.
So the mixing in Eq. 9.151 is the diagonalization of the gauge-boson mass matrix produced by the Higgs VEV. The angle θ_W is the angle that diagonalizes that matrix. Bonus prediction: $m_W = m_Z \cos\theta_W$ (Griffiths Eq. 9.98).

Final Summary

The Relation

The two relations (the answer to H2):

$$\begin{aligned} W^\pm_\mu &= \tfrac{1}{\sqrt{2}}\left(W^1_\mu \pm i\, W^2_\mu\right) && \text{(Griffiths Eq. 9.148)},\\ A_\mu &= B_\mu \cos\theta_W + W^3_\mu \sin\theta_W && \text{(Griffiths Eq. 9.151)},\\ Z_\mu &= -B_\mu \sin\theta_W + W^3_\mu \cos\theta_W && \text{(Griffiths Eq. 9.151)}, \end{aligned}$$

with the consistency relation $g_W \sin\theta_W = g' \cos\theta_W = g_e$, equivalently $\tan\theta_W = g'/g_W$ (**Griffiths Eq. 9.154**).
Why each holds:

- The $W^\pm$ relation is forced by (a) the requirement that the physical fields be complex, charge-conjugate pairs and (b) reproduction of the V-A vertex factor.
- The A, Z relation is forced by (a) the kinematic requirement of an orthogonal rotation to preserve canonical normalization and (b) reproduction of QED with coupling g_e at low energies, which fixes $\tan\theta_W = g'/g_W$.

The dynamical mechanism behind the A, Z mixing is the Higgs mechanism (Griffiths §10.9): the Higgs VEV gives mass to three of the four gauge fields, and the rotation in Eq. 9.151 is the diagonalization of the resulting gauge-boson mass matrix.